

Contoso Electronics

Equipment Reference & Maintenance Guide

X-Ray Inspection System

Contoso XView XR-900

Asset Identification

Field	Detail
Asset ID	CE-XRI-3400
Category	Quality Inspection
Manufacturer	Contoso Vision Systems
Location	Quality Lab - Room 4 (Shielded)
Installation Date	2023-07-08
Warranty Expiry	2028-07-08
Support Contact	quality.support@contoso.com ext. 4620

Overview

The Contoso XView XR-900 is an X-ray inspection system used to examine hidden solder joints such as BGAs, QFNs, and internal structures for voids, bridging, and alignment defects that optical inspection cannot detect.

Technical Specifications

Parameter	Value
Tube Voltage	Up to 130 kV
Focal Spot	1 µm (nanofocus)
Magnification	Up to 2000x geometric
Detector	Flat-panel digital
Sample Manipulator	5-axis, tilt to 70°
Radiation Safety	Fully shielded cabinet
Power	230 VAC, 50/60 Hz, 2 kW

Preventive Maintenance Schedule

Interval	Task
Daily	Verify interlocks and warning lamps function.
Weekly	Clean sample stage; check detector calibration image.
Monthly	Perform tube conditioning/warm-up cycle.
Annual	Radiation survey by certified officer; detector calibration.

Safety Guidelines

- Ionizing radiation - never bypass door interlocks or shielding.
- Only trained, authorized operators may run the system.
- Wear personal dosimeter as required by radiation safety program.
- Report any interlock fault immediately and stop use.

Common Issues & Troubleshooting

Symptom	Recommended Action
No X-ray image	Verify tube warm-up complete; check interlocks and tube status.
Poor image contrast	Adjust kV/current; run detector gain calibration.
Stage not moving	Check manipulator drives and limit switches.
Interlock fault	Do not override - inspect door switches; call service.

Contoso Electronics internal maintenance documentation. For service requests, contact quality.support@contoso.com | ext. 4620 or submit a work order through the maintenance portal.